ModelIt!

Student Activity Pack

What Happens When You Push Harder?

Force, Speed, and How Things Move

Kindergarten | NGSS K-PS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Circle the ball that was pushed harder: (picture of a fast-rolling ball with speed lines / picture of a slowly rolling ball)**

2. **Draw yourself pushing a ball really hard.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Push Force | Ball Speed

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gentle Tap

Push Force set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Power Push

Push Force set to high

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens to a ball's speed when you push it harder? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design a Speed Test Ramp

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YOUR ENGINEERING MISSION

Build a ramp and test how different push strengths make a ball travel at different speeds down the ramp.

THE SCENARIO

The school is building a new bowling alley game for Fun Day! Your team needs to figure out the PERFECT push to knock down the pins. Too soft and the ball is too slow. Too hard and it zooms past! Use what you learned about force and speed to design the winning push!

Engineering Design Thinking

1. HOW WILL YOU LINE UP THE PINS SO THE BALL CAN KNOCK THEM DOWN?
2. HOW WILL YOU MAKE SURE YOU PUSH THE SAME WAY EACH TIME?
3. HOW WILL YOU KEEP TRACK OF WHICH PUSH STRENGTH WORKED BEST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sports Scientists study how athletes throw, kick, and push to go faster and stronger. They help athletes win medals by understanding force and speed! They earn $50,000-$90,000/year.